In the name of Allah, the Most Beneficent, the Most Merciful

ECI671 Artificial Intelligence

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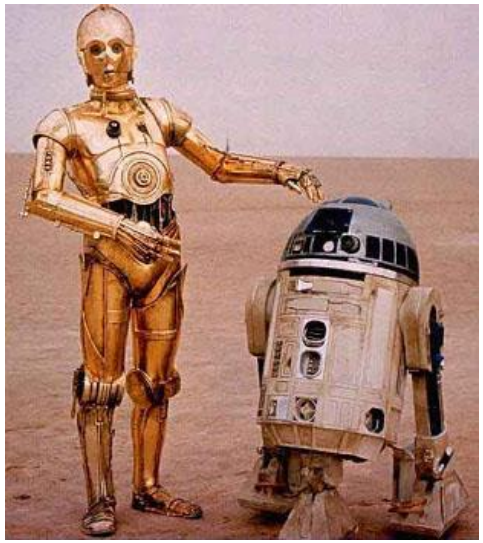
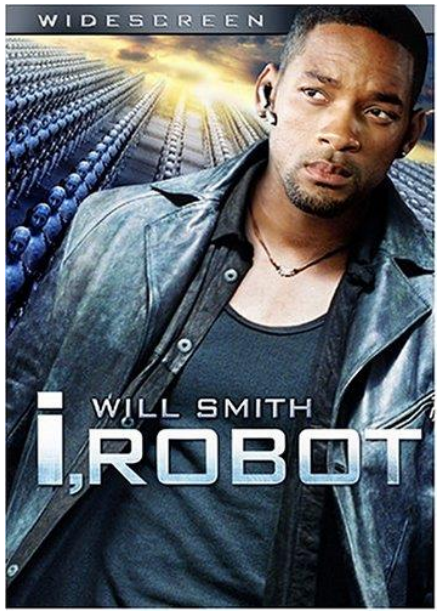
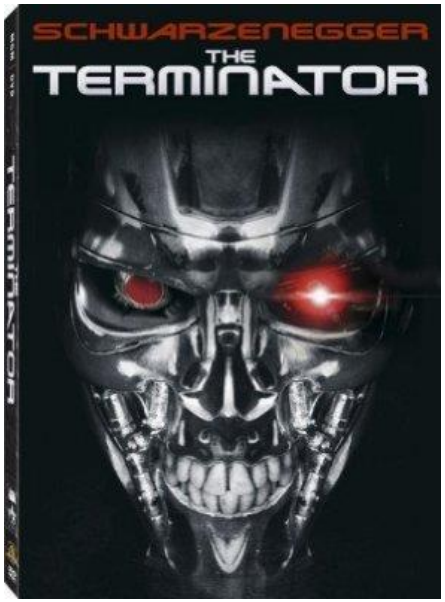
Outline

- What is AI
  - Definitions
  - Theories/hypotheses
- How it is relevant
- Impact on other areas

Artificial Intelligence in the Movies



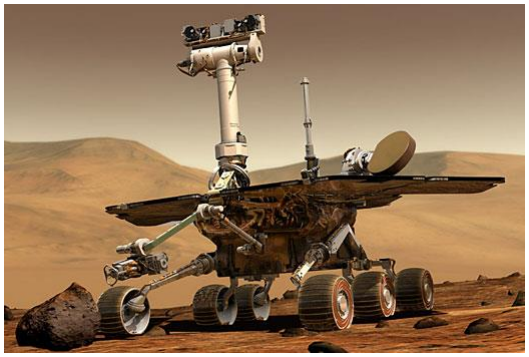


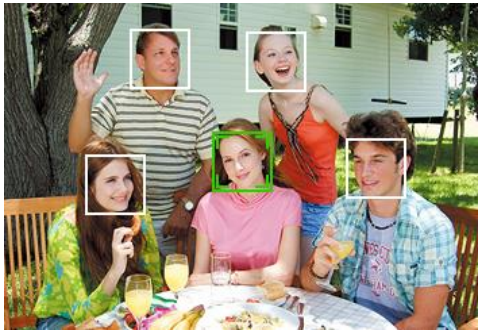


Artificial Intelligence in Real Life

A young *science* (≈ 50 years old)

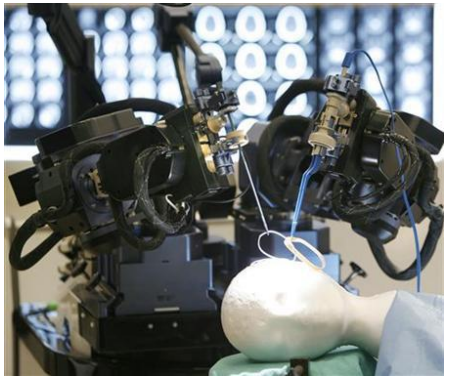
- Exciting and dynamic field, lots of uncharted territory left
- Impressive success stories
- "Intelligent" in specialized domains
- Many application areas





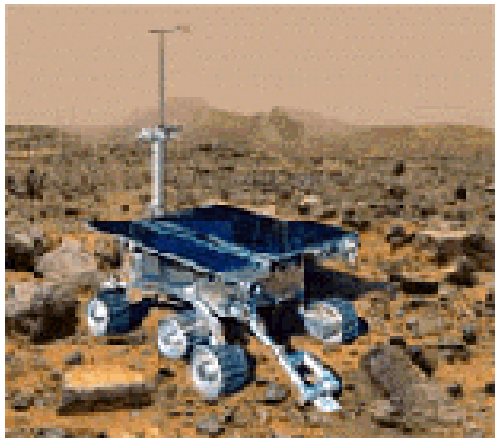
Face detection

Formal verification



Why the interest in AI?

Labor



Science

Search engines

Medicine/
Diagnosis

Appliances

What else?

Functions
Corporate Finance
Economic Studies
Governance
Information Technology
applications
management
networking
Marketing
Operations
Organization
Strategy

Industries
Automotive
Energy, Resources, Materials
Financial Services
Food & Agriculture

Information Technology: **Management**

Chart Pack

P PREMIUM

Article at a glance:

The return of **artificial intelligence**

Will artificial intelligence ever have any real applications in the business world? Those who have followed the cycles of hype around the technology during the past 20 years can be forgiven their skepticism. Now, however, the AI-development community has generated techniques that are beginning to show promise for solving real business problems involving complex data in dynamic environments—problems such as detecting fraud and automating work flows within and across organizations.

The take-away

Is artificial intelligence right for your business? The technology isn't appropriate for all information problems—but it does solve some of them very well indeed. This article suggests a three-step process for determining whether AI can help your company.

What is artificial intelligence?

- There is no clear consensus on the definition of AI
- John McCarthy coined the phrase AI in 1956

http://www.formal.stanford.edu/jmc/whatisai/whatisai.html

Q. What is artificial intelligence?

A. It is the science and engineering of making intelligent machines, especially intelligent computer programs. It is related to the similar task of using computers to understand human or *other* intelligence, but AI does not have to confine itself to methods that are biologically observable.

Q. Yes, but what is intelligence?

A. Intelligence is the computational part of the ability to achieve goals in the world. Varying kinds and degrees of intelligence occur in people, many animals and some machines.

What is AI? (Cont'd)

"The exciting new effort to make computers think ... *machines with minds*, in the full and literal sense" (Haugeland, 1985) "[The automation of] activities that we associate with human thinking, activities such as decision-making, problem solving, learning ..." (Bellman, 1978)	"The study of mental faculties through the use of computational models" (Charniak and McDermott, 1985) "The study of the computations that make it possible to perceive, reason, and act" (Winston, 1992)
"The art of creating machines that perform functions that require intelligence when performed by people" (Kurzweil, 1990) "The study of how to make computers do things at which, at the moment, people are better" (Rich and Knight, 1991)	"A field of study that seeks to explain and emulate intelligent behavior in terms of computational processes" (Schalkoff, 1990) "The branch of computer science that is concerned with the automation of intelligent behavior" (Luger and Stubblefield, 1993)

Figure 1.1 Some definitions of AI. They are organized into four categories:

Systems that think like humans.	Systems that think rationally.
Systems that act like humans.	Systems that act rationally.

Other possible AI definitions

- AI is a collection of hard problems which can be solved by humans and other living things, but for which we don't have good algorithms for solving.
  - e. g., understanding spoken natural language, medical diagnosis, circuit design, learning, self-adaptation, reasoning, chess playing, proving math theories, etc.
- Russsell & Norvig: a program that
  - Acts like human (Turing test)
  - Thinks like human (human-like patterns of thinking steps)
  - Acts or thinks rationally (logically, correctly)
- Some problems used to be thought of as AI but are now considered not
  - e. g., compiling Fortran in 1955, symbolic mathematics in 1965, pattern recognition in 1970, what for the future?

What is the scientific method hypothesis behind AI?

One Working Definition of AI

Artificial intelligence is the study of how to make computers do things that people are better at or would be better at if:

- they could extend what they do to a World Wide Web-sized amount of data and
- not make mistakes.

AI Purposes

"AI can have two purposes. One is to use the power of computers to augment human thinking, just as we use motors to augment human or horse power. Robotics and expert systems are major branches of that. The other is to use a computer's artificial intelligence to understand how humans think. In a humanoid way. If you test your programs not merely by what they can accomplish, but how they accomplish it, they you're really doing cognitive science; you're using AI to understand the human mind."

- Herb Simon

What's easy and what's hard?

- It's been easier to mechanize many of the high level cognitive tasks we usually associate with "intelligence" in people
  - e. g., symbolic integration, proving theorems, playing chess, some aspect of medical diagnosis, etc.
- It's been very hard to mechanize tasks that animals can do easily
  - walking around without running into things
  - catching prey and avoiding predators
  - interpreting complex sensory information (visual, aural, …)
  - modeling the internal states of other animals from their behavior
  - working as a team (ants, bees)
- Is there a fundamental difference between the two categories?
- Why are some complex problems (e.g., solving differential equations, database operations) are not subjects of AI?

History of AI

- AI has roots in a number of scientific disciplines
  - computer science and engineering (hardware and software)
  - philosophy (rules of reasoning)
  - mathematics (logic, algorithms, optimization)
  - cognitive science and psychology (modeling high level human/animal thinking)
  - neural science (model low level human/animal brain activity)
  - linguistics
- The birth of AI (1943 – 1956)
  - McCulloch and Pitts (1943): simplified mathematical model of neurons (resting/firing states) can realize all propositional logic primitives (can compute all Turing computable functions)
  - Alan Turing: Turing machine and Turing test (1950)
  - Claude Shannon: information theory; possibility of chess playing computers
  - Boole, Aristotle, Euclid (logics, syllogisms)

- Early enthusiasm (1952 – 1969)
  - 1956 Dartmouth conference
    John McCarthy (Lisp);
    Marvin Minsky (first neural network machine);
    Alan Newell and Herbert Simon (GPS);
  - Emphasis on intelligent general problem solving
    GSP (means-ends analysis);
    Lisp (AI programming language);
    Resolution by John Robinson (basis for automatic theorem proving);
    heuristic search (A*, AO*, game tree search)
- Emphasis on knowledge (1966 – 1974)
  - domain specific knowledge is the key to overcome existing difficulties
  - knowledge representation (KR) paradigms
  - declarative vs. procedural representation

- Knowledge-based systems (1969 – 1999)
  - DENDRAL: the first knowledge intensive system (determining 3D structures of complex chemical compounds)
  - MYCIN: first rule-based expert system (containing 450 rules for diagnosing blood infectious diseases)
    EMYCIN: an ES shell
  - PROSPECTOR: first knowledge-based system that made significant profit (geological ES for mineral deposits)
- AI became an industry (1980 – 1989)
  - wide applications in various domains
  - commercially available tools
  - AI winter
- Current trends (1990 – present)
  - more realistic goals
  - more practical (application oriented)
  - distributed AI and intelligent software agents
  - resurgence of natural computation - neural networks and emergence of genetic algorithms – many applications
  - dominance of machine learning (big apps)

AI is Controversial

- AI Winter – too much promised
- 1966: the failure of machine translation,
- 1970: the abandonment of connectionism,
- 1971−75: DARPA's frustration with the Speech Understanding Research program at Carnegie Mellon University
- 1973: the large decrease in AI research in the United Kingdom in response to the Lighthill report,
- 1973−74: DARPA's cutbacks to academic AI research in general,
- 1987: the collapse of the Lisp machine market,
- 1988: the cancellation of new spending on AI by the Strategic Computing Initiative
- 1993: expert systems slowly reaching the bottom
- 1990s: the quiet disappearance of the fifth-generation computer project's original goals,
- AI will cause
  - social ills, unemployment
  - End of humanity

Thinking Humanly: Cognitive Science

- 1960 “Cognitive Revolution”: information-processing psychology replaced behaviorism

- Cognitive science brings together theories and experimental evidence to model internal activities of the brain
  - What level of abstraction? “Knowledge” or “Circuits”?
  - How to validate models?
    - Predicting and testing behavior of human subjects (top-down)
    - Direct identification from neurological data (bottom-up)
    - Building computer/machine simulated models and reproduce results (simulation)

Thinking Rationally: Laws of Thought

- Aristotle (~ 450 B.C.) attempted to codify "right thinking"
  What are correct arguments/thought processes?
- E.g., "Socrates is a man, all men are mortal; therefore Socrates is mortal"

- Several Greek schools developed various forms of logic:
  <u>notation plus rules of derivation for thoughts.</u>

- <u>Problems:</u>
  1) Uncertainty: Not all facts are certain (e.g., *the flight might be delayed).*
  2) Resource limitations: There is a difference between solving a problem in principle and solving it in practice under various resource limitations such as time, computation, accuracy etc. (e.g., *purchasing a car)*

Strong vs Weak AI

- Strong AI possesses human-like intelligence, self-awareness, and consciousness, while weak AI is designed to perform a specific set of tasks within a narrow domain and does not possess these characteristics.

Strong AI

"I find it useful to distinguish what I will call "strong" AI from "weak" or "cautious" AI. According to weak AI, the principle value of the computer in the study of the mind is that it gives us a very powerful tool. For example, it enables us to formulate and test hypothesis in a more rigourous and precise fashion. But according to strong AI, the computer is not merely a tool in the study of the mind; rather, the appropriately programmed computer really *is* a mind, in the sense that computers given the right programs can be literally said to *understand* and have other cognitive states."

[Searle, 1980, *Minds, Brains and Programs*]

Weak and Strong AI Claims

- Weak AI:
  - Machines can be made to act *as if* they were intelligent.
- Strong AI:
  - Machines that act intelligently have real, conscious minds.

Eliza

- Called after Eliza Doolittle of Pygmalion fame.
- Developed in 1964-1966 by Joseph Weizenbaum in MIT
- Models (parodies) the rôle of a Rogerian psychotherapist engaged in an initial interview with a patient. Much or the technique of the Rogerian psychotherapist involves drawing the patient out by reflecting the patient's statements back at him.

https://web.njit.edu/~ronkowit/eliza.html

Eliza

Men are all alike
IN WHAT WAY
They’re always bugging us about something or other.
CAN YOU THINK OF A SPECIFIC EXAMPLE
Well, my boyfriend made me come here.
YOUR BOYFRIEND MADE YOU COME HERE
He says I am depressed much of the time.
I AM SORRY TO HEAR YOU ARE DEPRESSED
It’s true. I am unhappy.
DO YOU THINK COMING HERE WILL HELP YOU NOT TO BE UNHAPPY
I need some help, that much seems certain.
WHAT WOULD IT MEAN TO YOU IF YOU GOT SOME HELP
Perhaps I could learn to get along with my mother.

Eliza

TELL ME MORE ABOUT YOUR FAMILY
My mother takes care of me.
WHO ELSE IN YOUR FAMILY TAKES CARE OF YOU
My father.
YOUR FATHER
You are like my father in some ways
WHAT RESEMBLANCE DO YOU SEE
You are not very aggressive but I think you don’t want me to notice that.
WHAT MAKES YOU THINK I AM NOT VERY AGGRESSIVE
You don’t argue with me
WHY DO YOU THINK I DON’T ARGUE WITH YOU

What is/was Eliza?

- A chatterbot
- Turing test competitor
- A friend
- A model for other bots

ELIZA

- ELIZA simulated conversation by using a pattern matching and substitution methodology that gave users an illusion of understanding on the part of the program, but had no representation that could be considered really understanding what was being said by either party. Whereas the ELIZA program itself was written (originally) in MAD-SLIP, the pattern matching directives that contained most of its language capability were provided in separate "scripts", represented in a lisp-like representation.

What is a chatterbot?

- A chatter robot, chatterbot, chatbot, or chat bot is a computer program designed to simulate an intelligent conversation with one or more human users via auditory or textual methods, primarily for engaging in small talk.
  - The primary aim of such simulation has been to fool the user into thinking that the program's output has been produced by a human (the Turing test).
  - Programs playing this role are sometimes referred to as Artificial Conversational Entities, talk bots or chatterboxes.
  - Uses:
    - chatterbots are often integrated into dialog systems for various practical purposes such as online help, personalised service, or information acquisition.
    - Spam in chatrooms
  - Some chatterbots use sophisticated natural language processing systems, but many simply scan for keywords within the input and pull a reply with the most matching keywords, or the most similar wording pattern, from a textual database.
  - Collections:

http://www.simonlaven.com/

Types of Chatterbots

- Classic Chatterbots
- Complex Chatterbots
- Friendly Chatterbots
- Teachable Bots
- AIML Bots
- JFred Bots
- NativeMinds Bots Non-English Bots
- Alternative Bots

http://www.simonlaven.com/

A.L.I.C.E

ALICEBOT

ALICE A.I. FOUNDATION NEWS ITEMS, ANNOUNCEMENTS, PRESS RELEASES AND OPINIONS. ANYTHING RELATED TO ARTIFICIAL INTELLIGENCE MARKUP LANGUAGE (AIML), THE TURING TEST, THE LOEBNER PRIZE, BOT HOSTING SERVICES, FREE AND PROPRIETARY CHAT BOT SOFTWARE.

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Dr. Richard Wallace
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Virtual DJ Denise is the First Artificial Intelligent Radio Host http://t.co/A7sSNHh
8 days ago · reply · retweet · favorite

Robot teaches English as Second Language http://t.co/6MpwQip
10 days ago · reply · retweet · favorite

Superbot Offer – Free Consulting http://t.co/NEC9ceT
44 days ago · reply · retweet · favorite

Emma the Cybrarian: http://t.co/elWJMZr
57 days ago · reply · retweet · favorite

ALICE, Dick Cavett, R.L. Stine and me on the WFMU radio show 7 Second Delay http://t.co/YyPhFOB
85 days ago · reply · retweet · favorite

twitter

WEDNESDAY, JULY 27, 2011

AIML Superbot Special Offer - Free Consulting

The AIML Superbot is a blank template that helps you create a custom chat bot from scratch. If you want to create a bot with its own proprietary, unique personality, the Superbot helps you by providing a rich set of patterns that allow you to "fill in the blanks" with your own bot responses.

Building a bot from scratch can still be a daunting task for those new to AIML and chat bot creation. That is why we are now offering 2 hours of free consulting time with your Superbot purchase. You will be able to talk directly by phone or Skype with Dr. Richard S. Wallace or another AIML expert and receive support and help with your bot project.

Use your consultation time to get specific answers to all your questions about bot content creation and help with your project, including help linking the bot to speech recognition systems, avatars, speech synthesis and other third-party applications.

This offer expires on October 1, 2011.

POSTED BY DR. RICHARD S. WALLACE AT 8:37 AM 2 COMMENTS
LABELS: AIML, AVATARS, BOT HOSTING, CHATBOTS, CONSULTING, SUPERBOT, TEXT TO SPEECH

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Passion for Design? Turn Your Passion into a Profession at The Art
Top 10 HRIS Software 2011 Top 10 Human Resource Software Vend
Microsoft® Office 365 Access Email, Documents & Calendars Anytir

FRIDAY, MAY 27, 2011

New Release of ALICE AIML on Google Code

Click here to chat with Talking Animated Fake Captain Kirk

SUBSCRIPTION BOTS

A.L.I.C.E. Silver Edition
CLAUDIO Personality Test
DAVE E.S.L. Bot

MORE AIML RESOURCES

Pandorabots
Oddcast Avatars
Robot-Hosting
Riot Software

MORE BOT RESOURCES

Wikipedia
Agentland
Open Directory
Virtual Humans

KUKI

- Formerly Mitsuku
- Written by Steve Worswick
- Uses Pandorabots AIML technology, statistical models and machine learning
- Loebner Prize (in 2013, 2016, 2017, 2018, and 2019)
- Kuki.ai

Philosophical criticisms of AI

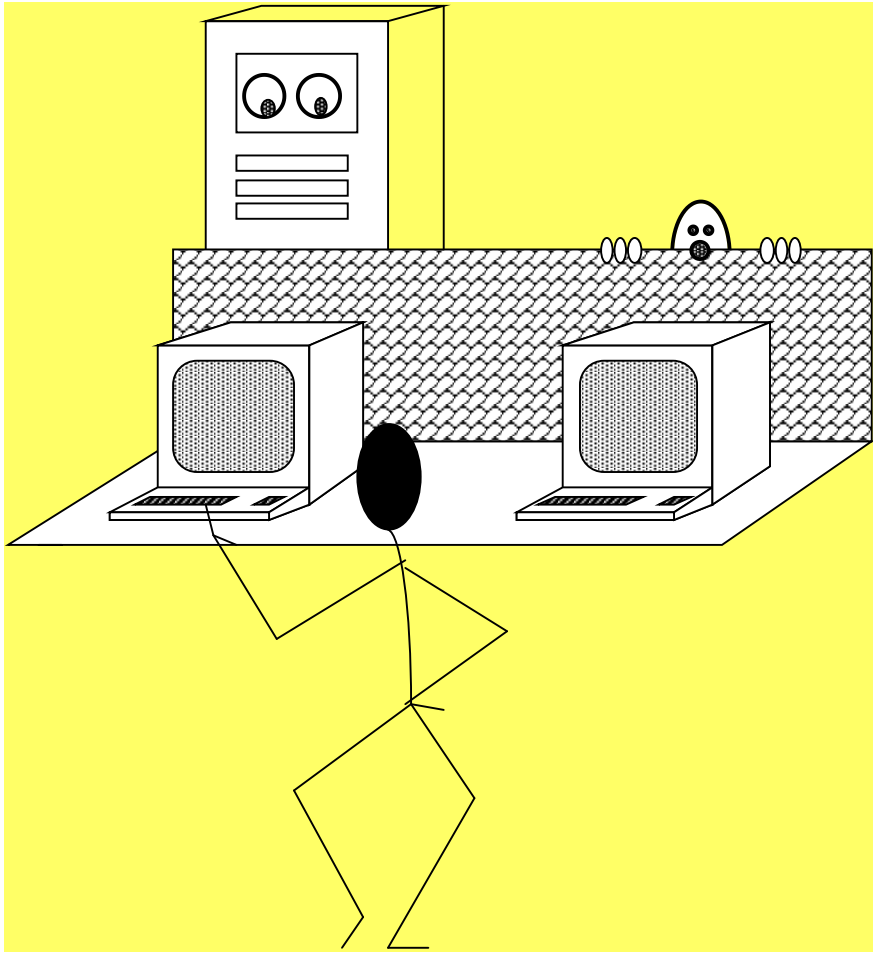
- Two categories of criticism:
  - It cannot be done because ...
  - It cannot be done the way you are trying to do it.

The danger of can't be done arguments…

"Philosophers are forever telling scientists what they can't do, what they can't say, what they can't know, and so on and so forth. In 1844 the philosopher August Compte said that if there was one thing man would never know it would be the composition of the distant stars and planets. But three years after Compte died physicists discovered that an object's composition can be determined by its spectrum no matter how far off the object happens to be."

What is Intelligence?

The Turing Test



A machine can be described as a thinking machine if it passes the Turing Test. i.e. If a human agent is engaged in two isolated dialogues (connected by teletype say); one with a computer, and the other with another human and the human agent cannot reliably identify which dialogue is with the computer.

Intelligence

- *Turing Test*: A human communicates with a computer via a teletype. If the human can't tell he is talking to a computer or another human, it passes.
  - Natural language processing
  - knowledge representation
  - automated reasoning
  - machine learning
- Add vision and robotics to get the total Turing test.

Turing Test – Loebner prize

Loebner Prize Judges Could Easily Identify Chatbots

Written by Sue Gee

Friday, 18 May 2012 08:36

The 2012 Loebner Prize for the best chatbot has been awarded to Chip Vivant, created by Mohan Embar, a software consultant based in Milwaukee.

There were four contestants in the final round of this competition which took place at Bletchley Park on May 15, 2012 and none of them were likely to fool the judges into mistaking them for a human as required to win the Loebner Prize Gold Medal. So instead the Bronze Medal and $5,000 was awarded to the chatbot with the most impressive conversational abilities.

Alan Turing depicted on the Loebner Prize Gold Medal

Objections to the TT

- **The Theological Objection**
  - "Thinking is a function of man's immortal soul. God has given an immortal soul to every man and woman, but not to any other animal or to machine. Hence no animal or machine can think."
- **The "Head in the Sand" Objection**
  - "The consequences of machines thinking are to dreadful to think about."

Objections to the TT

- **Mathematical Objections**
  - "There are a number of results of mathematical logic that can be used to show that there are limitations to the power of discrete state machines.“
    - (eg. Gödel’s incompleteness theorem)
- **The Argument for Consciousness**
  - “A machine cannot write a sonnet or compose a concerto because of thoughts or emotions felt.”

Types of Intelligence Tests

The Imitation Game

***Questioner*:** **Aims to discover if A or B is the Man**	**Questions** ----------> <--------- **Answers**	**(A) Male: aims to fool the questioner** **(B) Female: aims to help the questioner**

Turing's Test

***Questioner*:** **Aims to discover if A or B is the Computer**	**Questions** ----------> <--------- **Answers**	**(A) Computer: aims to fool the questioner** **(B) Human: aims to help the questioner**

Modified Turing Test Setup

***Questioner*:** **Aims to discover if the querent is human or computer.**	**Questions** ----------> <--------- **Answers**	**Human or Computer**

Turing's rationale: "The question and answer method seems to be suitable for introducing almost any one of the fields of human endeavor which we wish to include" (Turing 1950, p.435).

Turing's analysis: "The [imitation] game may perhaps be criticized on the ground that the odds are weighted too heavily against the machine. This objection is a very strong one, but at least we can say that if, nevertheless, a machine can be constructed to play the imitation game satisfactorily, we need not be troubled by this objection" (Turing 1950, p. 435).

Turing's prediction: "in about fifty years' time [by the year 2000] it will be possible to program computers ... to make them play the imitation game so well that an average interrogator will have no more than 70 per cent. chance of making the correct identification after five minutes of questioning." (Turing 1950, p.442).

Connectionist (Subsymbolic) Hypothesis

"The intuitive processor is a subconceptual connectionst dynamical system that does not admit a complete, formal and precise conceptual-level description." [Smolensky 1988]

The inner workings of an ANN are difficult to interpret – but are they substantially different to a symbolic system?

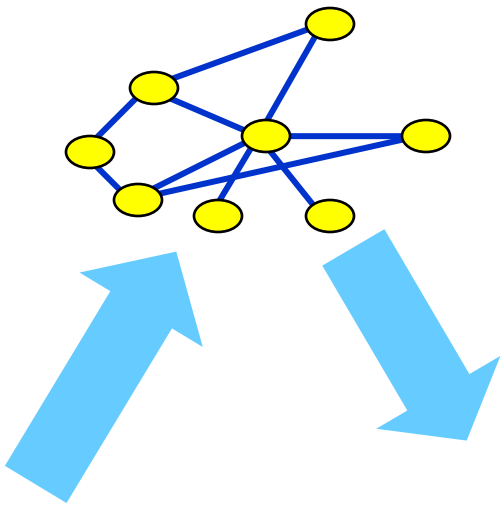
Physical Symbol System Hypothesis

- *A physical symbol system has the necessary and sufficient means for intelligent action*

  Newell & Simon 1976

  - a system, embodied physically, that is engaged in the manipulation of symbols
  - an entity is potentially intelligent if and only if it instantiates a physical symbol system
  - symbols must *designate*
  - symbols must be *atomic*
  - symbols may combine to form *expressions*

What does the PSSH mean?



- Intelligent action can be modelled by a system manipulating symbols.
- Nothing special about our *wetware*.
- Intelligence can be implemented on other platforms, e.g. silicon.

Symbolic AI: Rule-Based Systems

- Whale Watcher Demo
  - http://www.aiinc.ca/demos/whale.shtml

Symbolic AI: Rule-Based Systems

- Symbolic AI is the collection of all methods in artificial intelligence research that are based on high-level symbolic representations of problems, logic and search.
- It used tools such as
  - logic programming,
  - production rules,
  - semantic nets and
  - frames

Symbolic AI: Rule-Based Systems

- Symblic AI developed applications such as
  - knowledge-based systems (expert systems),
  - symbolic mathematics,
  - automated theorem provers,
  - ontologies,
  - the semantic web, and
  - automated planning and scheduling systems.
- Symbolic AI was the dominant paradigm of AI research from the mid-1950s until the mid-1990s

Rule-Based System: Car Maintenance

BadElecSys:

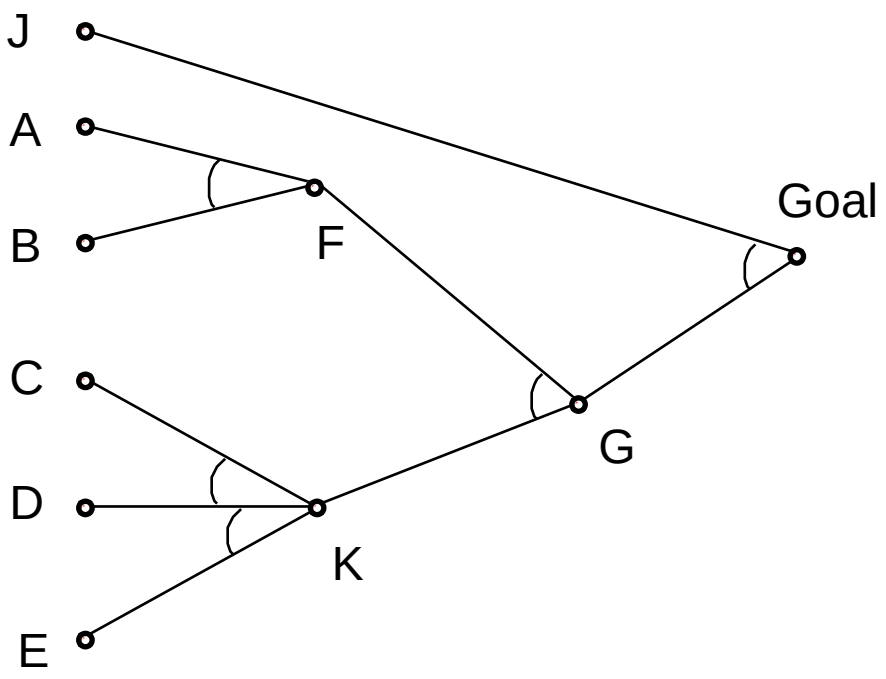
IF car:SparkPlusCondition #= Bad Or
car:Timing #= OutOfSynch Or
car:Battery #= Low;
THEN car:ElectricalSystem = Bad;

***GoodElecSys*:**

IF car:SparkPlugCondition #= Ok And
car:Timing #= InSynch And
car:Battery #= Charged;
THEN car:ElectricalSystem = Ok;

Consider the following rules

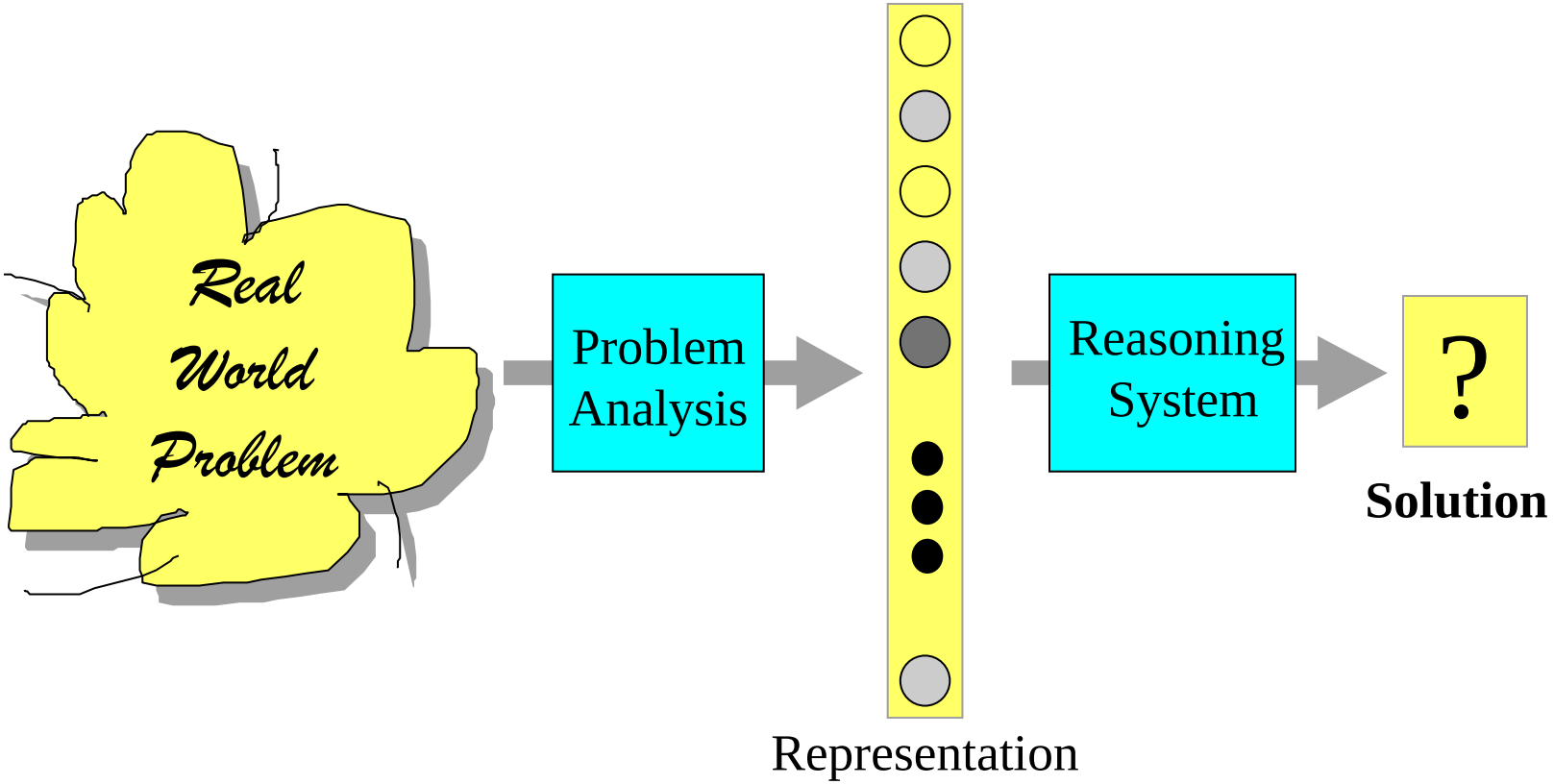
```
If A and B then F
If C and D
     and E then K
If F and K then G
If J and G then Goal
```



We can **Forward Chain** from Premises to Goals

or **Backward Chain** from Goals and try to prove them.

A model of knowledge-based systems development



http://www.loebner.net/Prizef/loebner-prize.html

Home Page of The Loebner Prize--"The First Turing Test"

Loebner Prize Gold Medal

(Solid 18 carat, not *gold-plated* like the Olympic *"Gold"* medals)

What is the Loebner Prize?

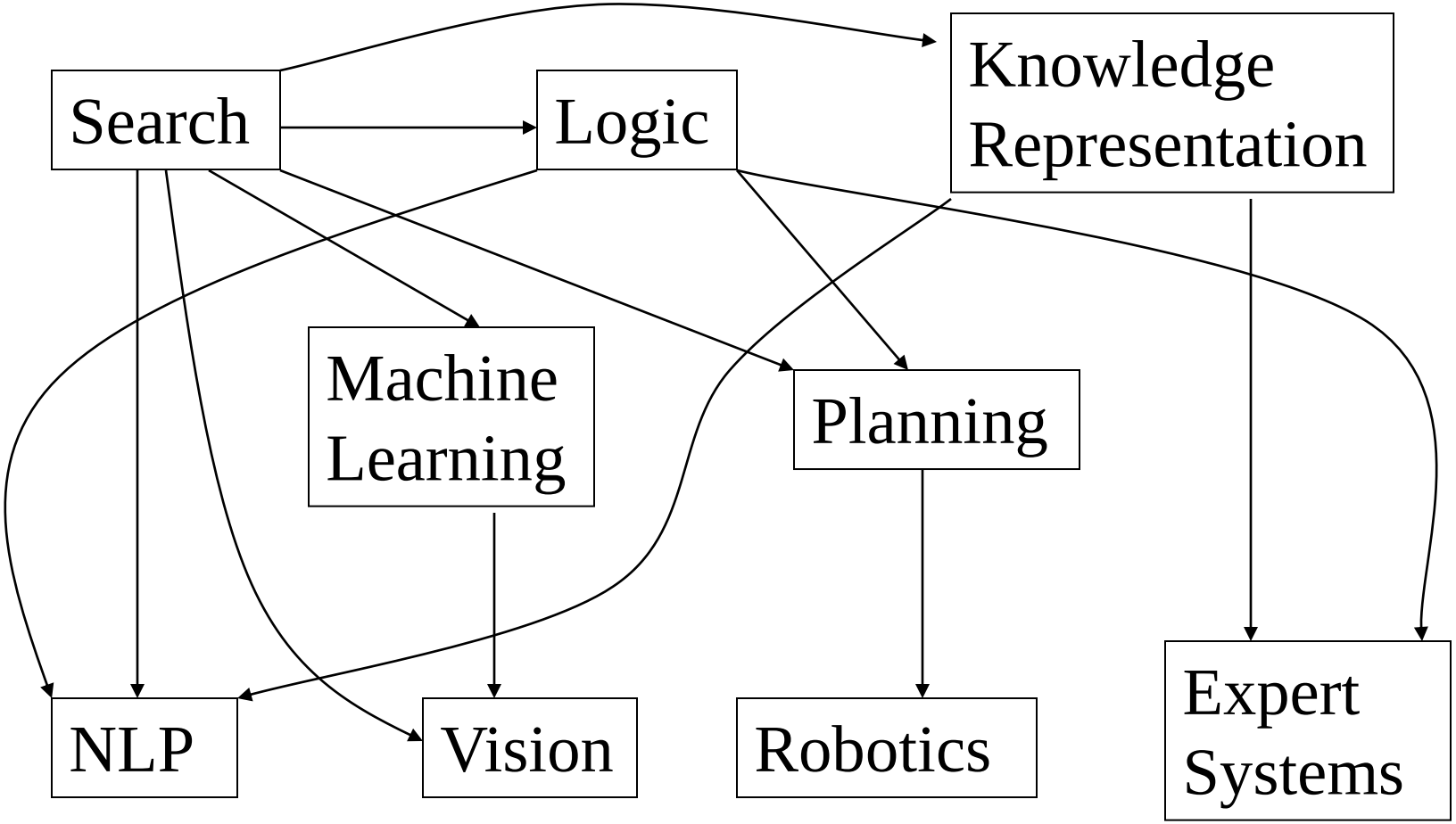
The Loebner Prize is the first formal instantiation of a Turing Test. The test is named after Alan Turing the brilliant British mathematician. Among his many accomplishments was basic research in computing science. In 1950, in the article *Computing Machinery and Intelligence* which appeared in the philosophical journal Mind, Alan Turing asked the question "Can a Machine Think?"He answered in the affirmative, but a central question was: "If a computer could think, how could we tell?" Turing's suggestion was, that if the responses from the computer were indistinguishable from that of a human,the computer could be said to be thinking.

In 1990 Hugh Loebner agreed with The Cambridge Center for Behavioral Studies to underwrite a contest designed to implement the Turing Test. Dr. Loebner pledged a Grand Prize of $100,000 and a Gold Medal (pictured above) for the first computer whose responses were indistinguishable from a human's. Each year an annual prize of $2000 and a bronze medal is awarded to the **most** human computer. The winner of the annual contest is the best entry relative to other entries that year, irrespective of how good it is in an absolute sense.

Branches of AI

- **Logical AI**
- **Search**
- **Natural language processing**
- **Computer vision**
- **Pattern recognition**
- **Knowledge representation**
- **Inference** From some facts, others can be inferred.
- **Reasoning**
- **Learning**
- **Planning** To generate a strategy for achieving some goal
- **Epistemology** This is a study of the kinds of knowledge that are required for solving problems in the world.
- **Ontology** Ontology is the study of the kinds of things that exist.
- **Agents**
- **Games**
- **Artificial life / worlds?**
- **Emotions?**
- **Knowledge Management?**
- **Socialization/communication?**
- …

Areas of AI and Some Dependencies



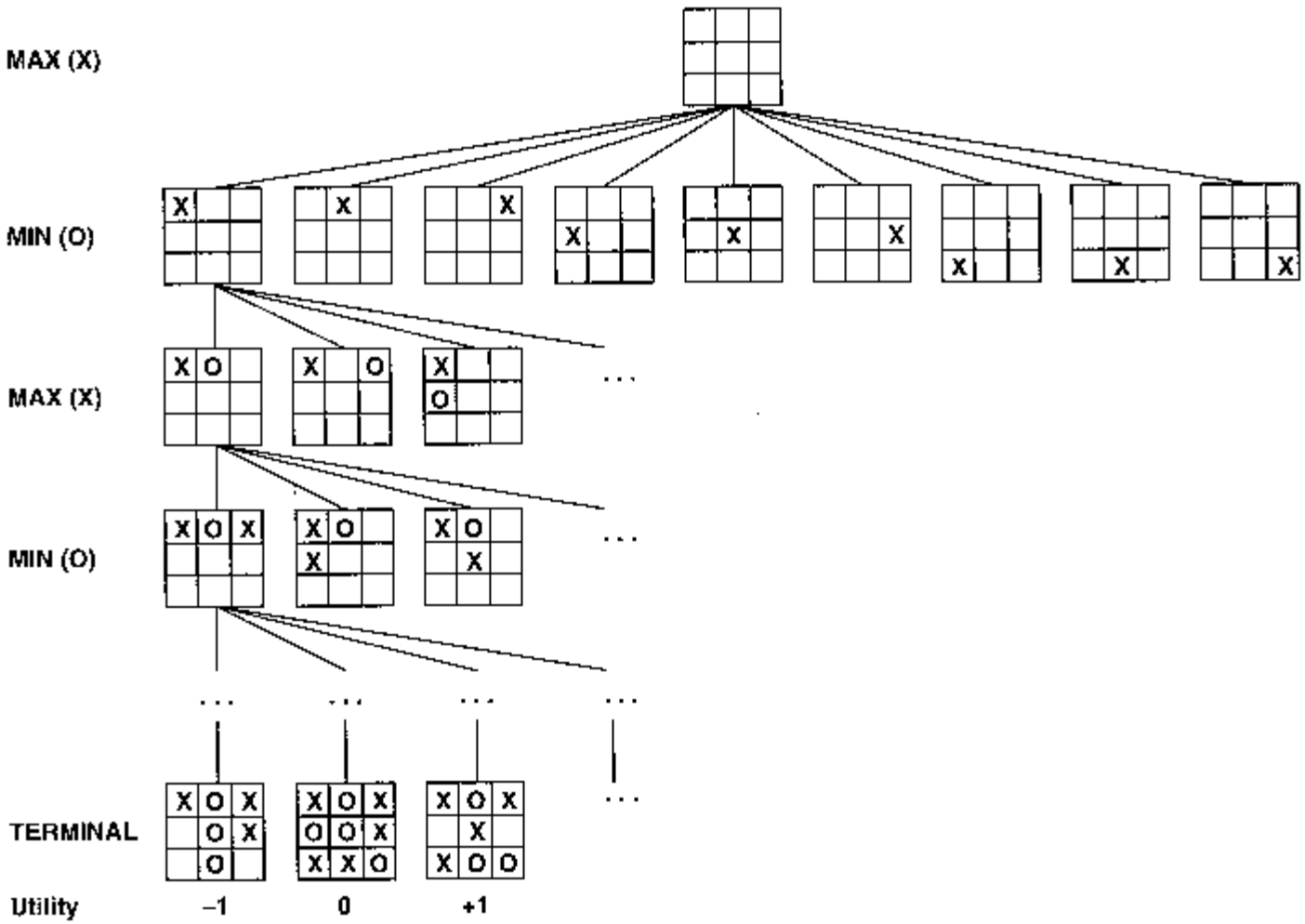
Approaches to AI

- Searching
- Learning
- From Natural to Artificial Systems
- Knowledge Representation and Reasoning
- Expert Systems and Planning
- Communication, Perception, Action

Search

- "All AI is search"
  - Game theory
  - Problem spaces
- Every problem is a <u>feature space</u> of all possible (successful or unsuccessful) solutions.
- The trick is to find an efficient search strategy.

Search: Game Theory



9!+1 = 362,880

Approaches to AI

- Searching
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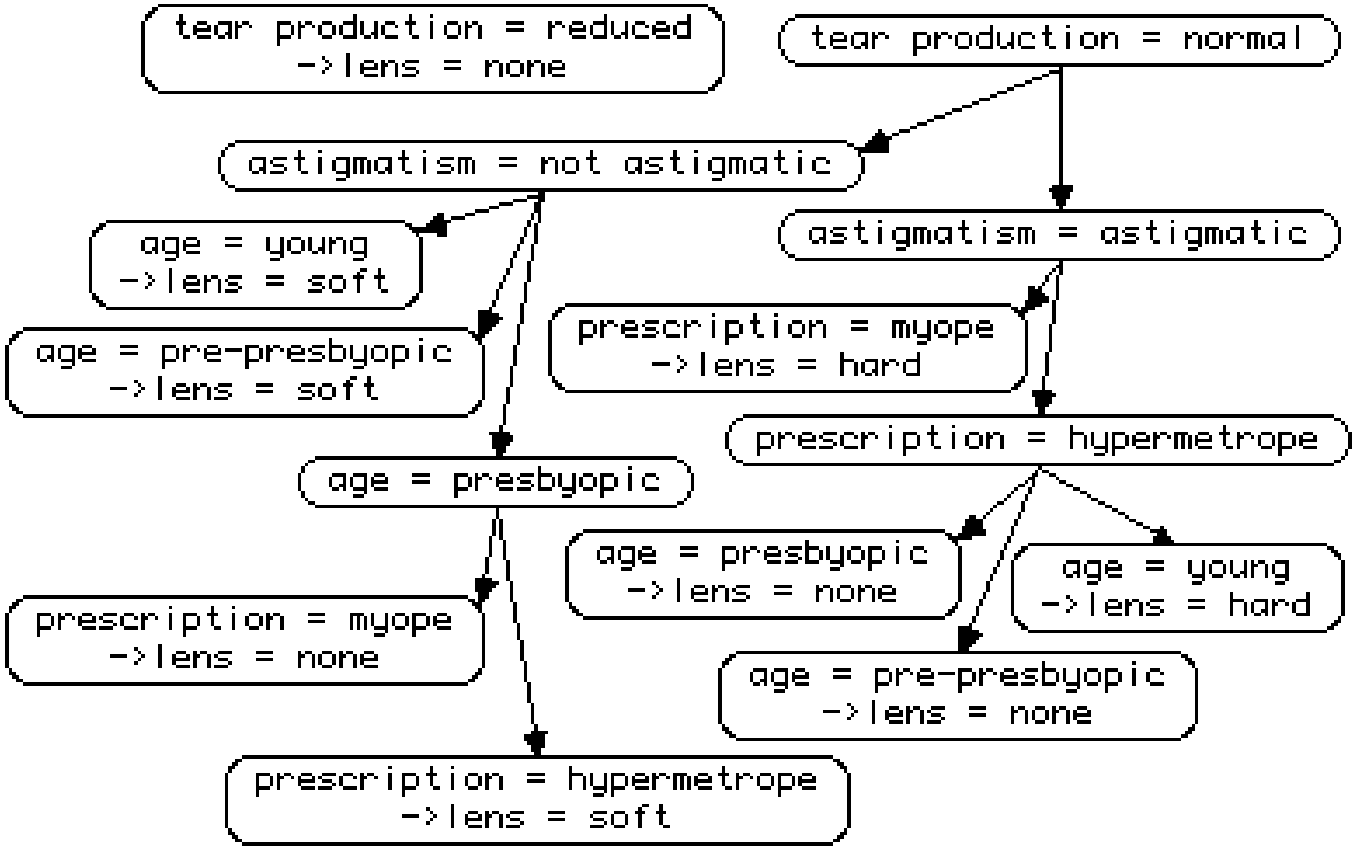
Learning

- Explanation
  - Discovery
  - Data Mining
- No Explanation
  - Neural Nets
  - Case Based Reasoning

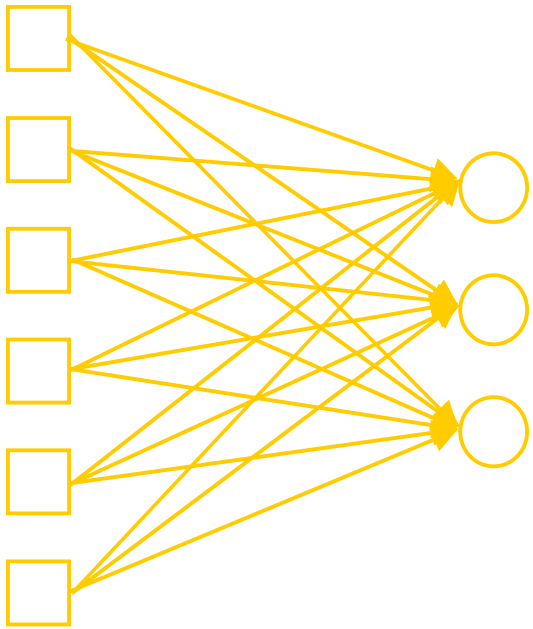
Learning: Explanation

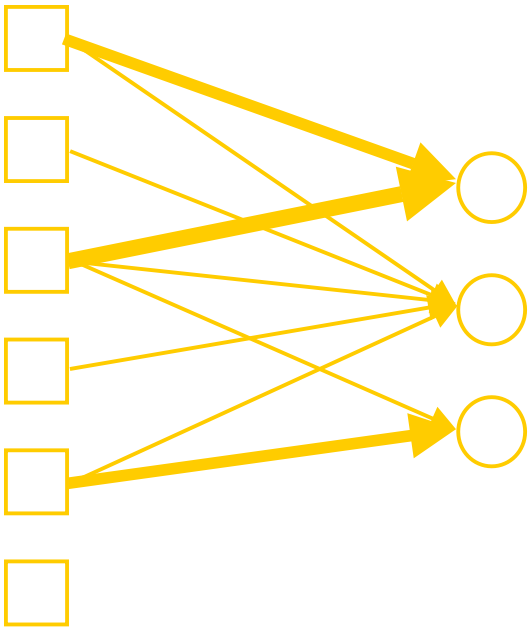
- Cases to rules



Learning: No Explanation

- Neural nets





Approaches to AI

- Searching
- Learning
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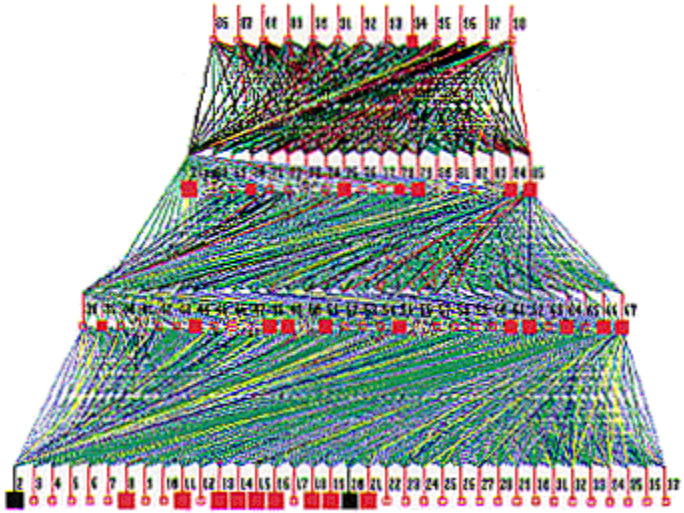
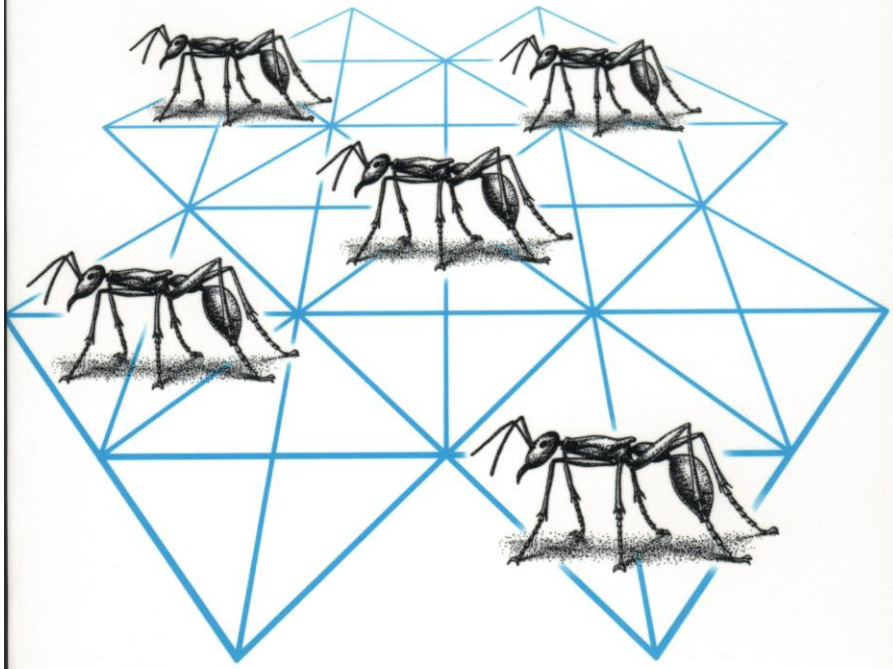
Neural Networks

→



Swarm Intelligence

From Natural to Artificial Systems



Eric Bonabeau
Marco Dorigo
Guy Theraulaz

A VOLUME IN THE
SANTA FE INSTITUTE STUDIES IN THE SCIENCES OF COMPLEXITY

Approaches to AI

- Searching
- Learning
- From Natural to Artificial Systems
- Knowledge Representation and Reasoning
- Expert Systems and Planning
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Rule-Based Systems

- Logic Languages
  - Prolog, Lisp
- Knowledge bases
- Inference engines

Rule-Based Languages: Prolog

Father(abraham, isaac).
Father(haran, lot).
Father(haran, milcah).
Father(haran, yiscah).

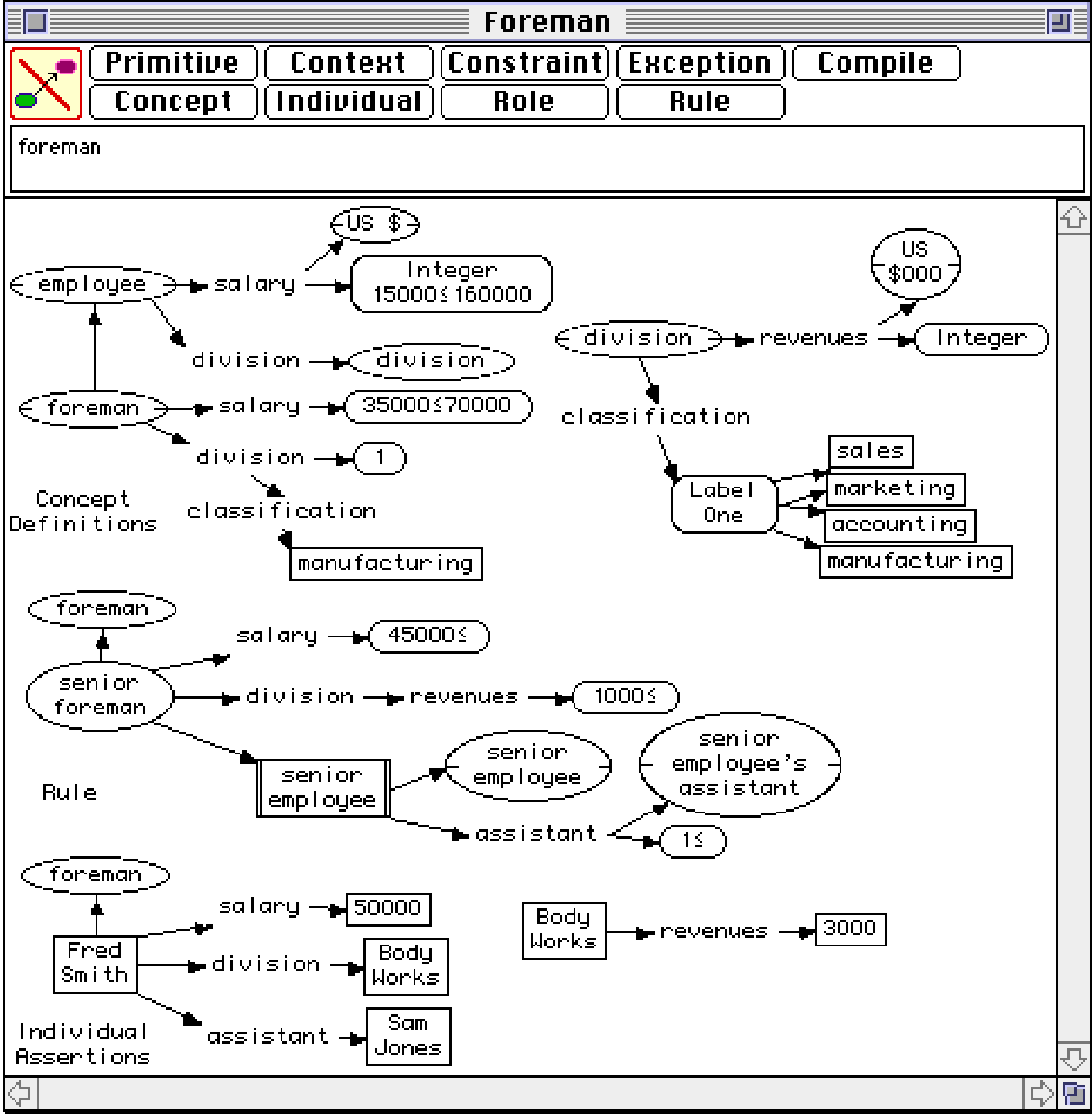
Male(isaac).
Male(lot).
Female(milcah).
Female(yiscah).

Son(X,Y) $\leftarrow$ Father(Y,X), Male(X).
Daughter(X,Y) $\leftarrow$ Father(Y,X), Female(X).

Son(lot, haran)?

Rule Based Systems

- KRS



Approaches to AI

- Searching
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Approaches to AI

- Searching
- Learning
- From Natural to Artificial Systems
- Knowledge Representation and Reasoning
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Ability-Based Areas

- Computer vision
- Natural language recognition
- Natural language generation
- Speech recognition
- Speech generation
- Robotics
- Games/entertainment

MIT’s NLP online

http://start.csail.mit.edu/

MIT’s NLP online

- The START Natural Language System is a software system designed to answer questions that are posed to it in natural language. START parses incoming questions, matches the queries created from the parse trees against its knowledge base and presents the appropriate information segments to the user. In this way, START provides untrained users with speedy access to knowledge that in many cases would take an expert some time to find.
- START (SynTactic Analysis using Reversible Transformations) was developed by Boris Katz at MIT's Artificial Intelligence Laboratory. Currently, the system is undergoing further development by the InfoLab Group, led by Boris Katz. START was first connected to the World Wide Web in December, 1993, and in its several forms has to date answered millions of questions from users around the world.

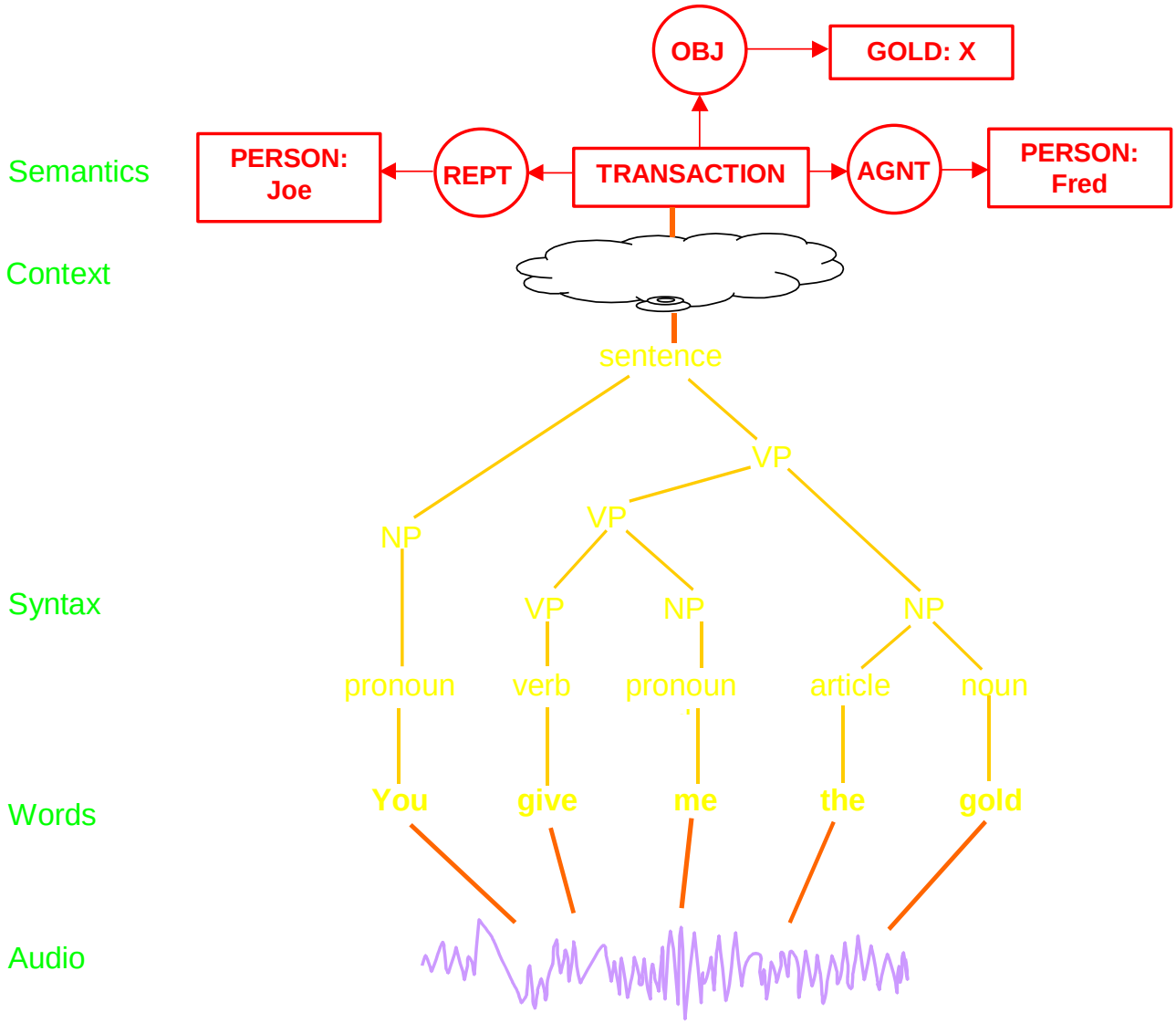
Natural Language: Translation

"The flesh is weak, but the spirit is strong"

→ Translate to Russian
→ Translate back to English

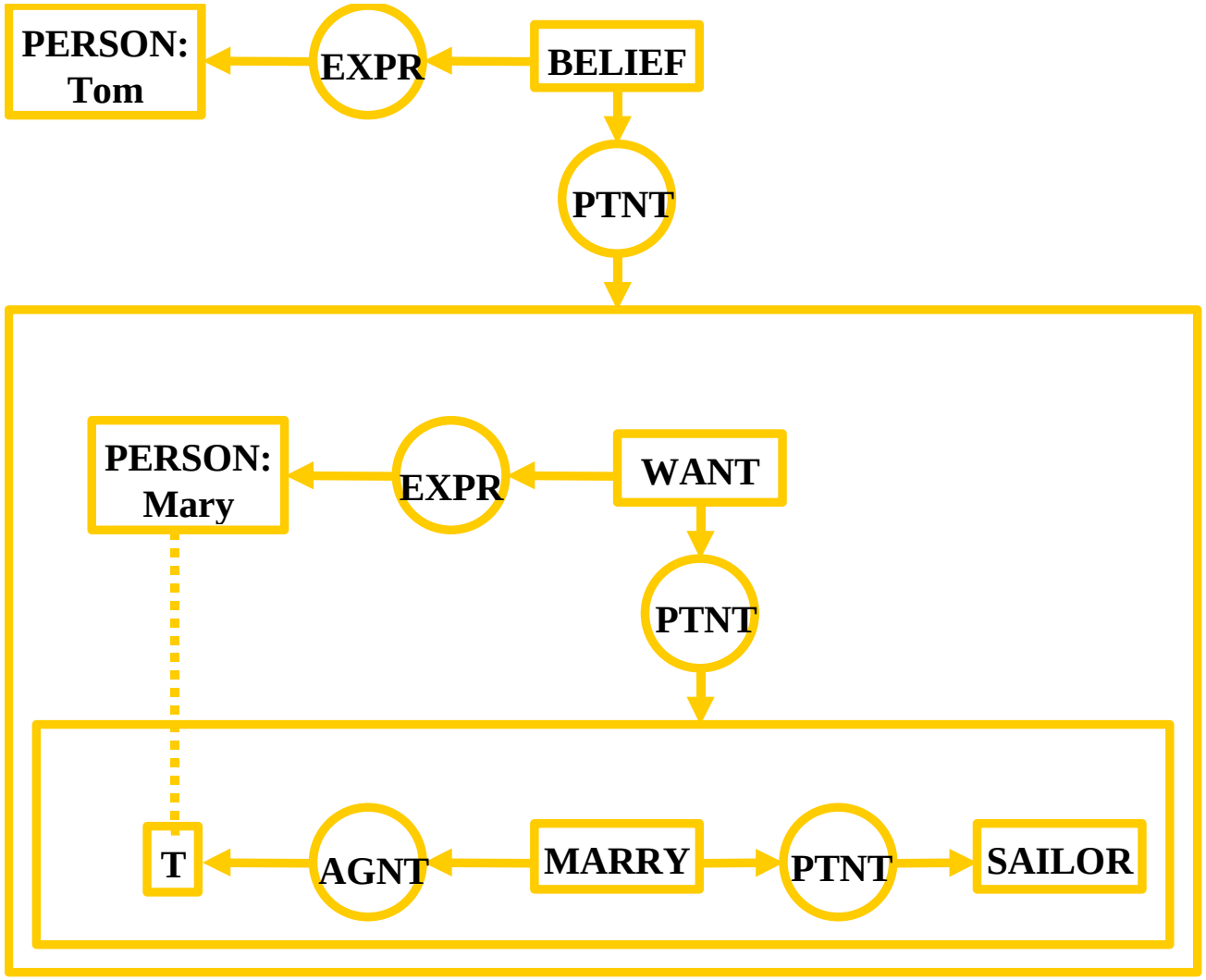
"The food was lousy, but the vodka was great!"

Natural Language Recognition

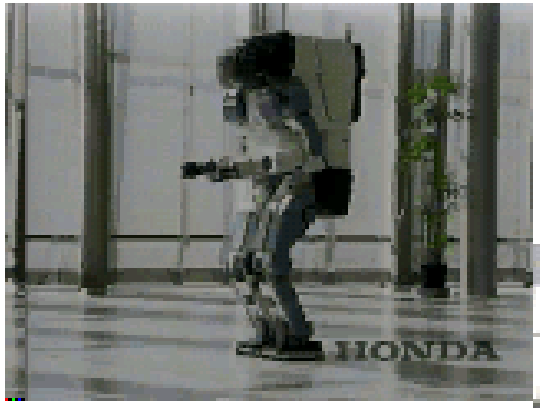


Natural Language Recognition

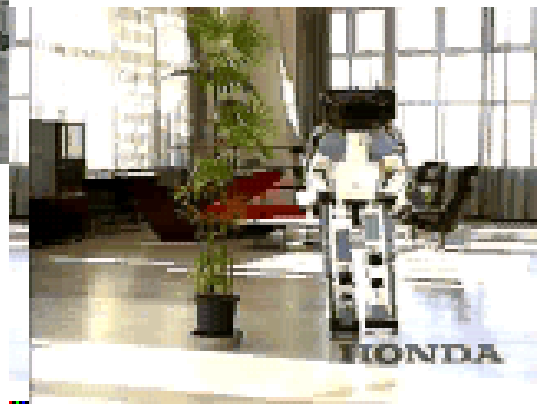
"Tom believes Mary wants to marry a sailor."



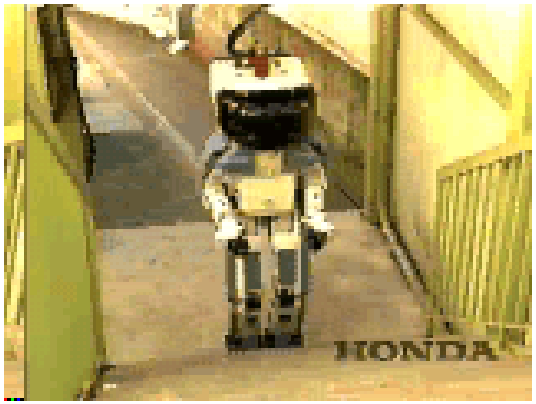
Honda Humanoid Robot



Walk



Turn



Stairs

Honda Humanoid Robot

ASIMO (**Advanced Step in Innovative Mobility**) is a humanoid robot created by Honda in 2000. It is displayed in the Miraikan museum in Tokyo, Japan. On 8 July 2018, Honda posted the last update of Asimo on their official page stating that it would be ceasing all development and production of Asimo robots in order to focus on more practical applications using the technology developed through Asimo's lifespan. It made its last active appearance in March 2022, over 20 years after its first, as Honda announced that they are retiring the robot to concentrate on remote-controlled, avatar-style, robotic technology.

Domestic Robots

Robotshop.com

Domestic Robot Accessories

Domestic Robot Replacement Parts

Robot Vacuums

Robot Floor Cleaners

Robot Pool Cleaners

Robotic Lawn Mowers | Lawnbott Robot Mower

Robot Pet Care

Robotic Window Cleaners

Sweeping Robots

RobotShop Packages

Robots for Sports

Surveillance and Companion Robots

Military robots

Your artificial intelligent companion

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Important announcement

Following the Sony Corporation FY05 3Q announcement, the production of AIBO Entertainment Robots has been discontinued as of end March 2006. As for the sales activity, we will discontinue the sales of AIBO once all remaining stock runs out.

Read more

Terms and conditions of web site ©2006 Sony Entertainment Robot Europe Privacy statement

like.no.other™ SONY

Robocup

www.robocup.org

How far have we got?

- General intelligence of a frog?

But then ask Garry K.

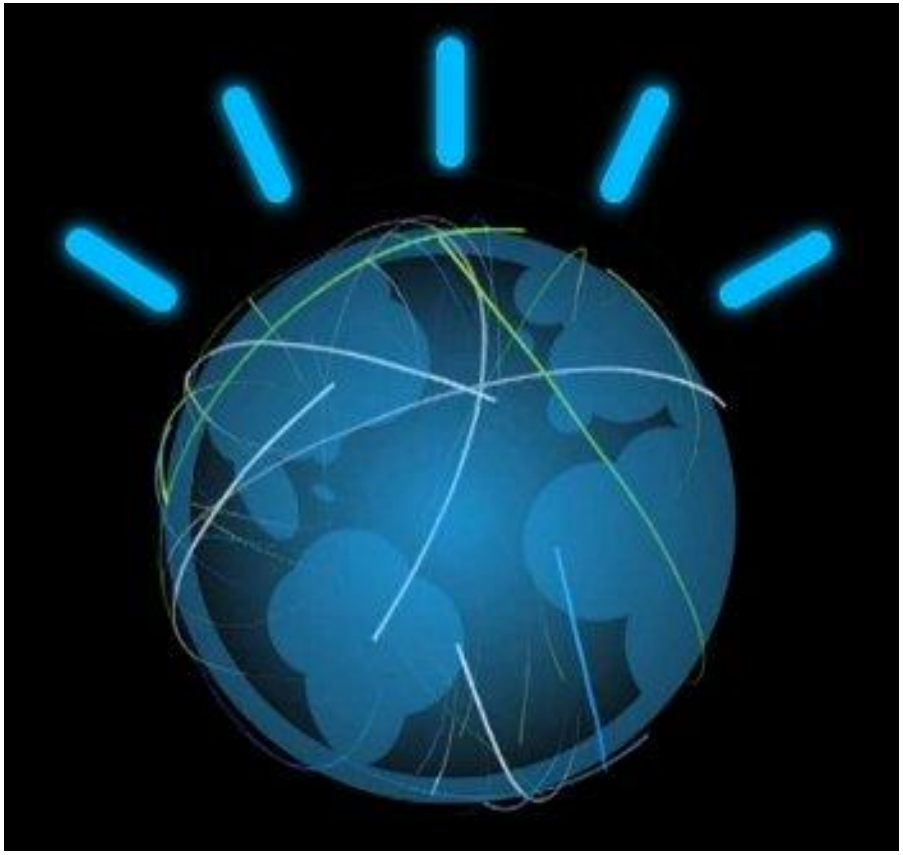
But don’t try to ask Deep Blue

Watson

- “The goal is to have computers start to interact in natural human terms across a range of applications and processes, understanding the questions that humans ask and providing answers that humans can understand and justify” - IBM

Watson

- IBM’s Artificial Intelligence computer system
- Capable of answering questions in natural language
- Competed against champions on Jeopardy and won

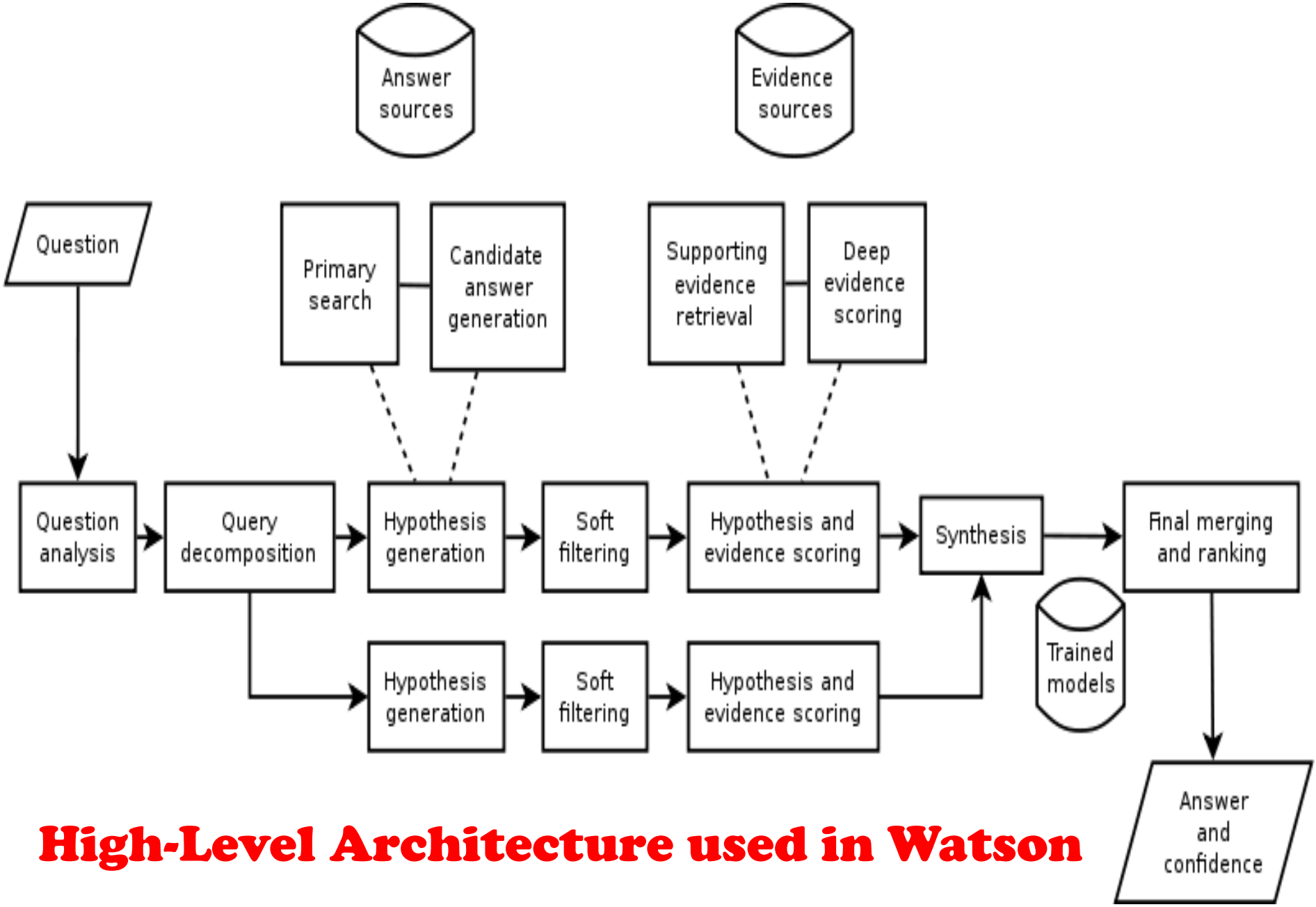


Watson

- IBM describes this AI as:

  "an application of advanced Natural Language Processing, Information Retrieval, Knowledge Representation and Reasoning, and Machine Learning technologies to the field of open domain question answering“

- What this means…

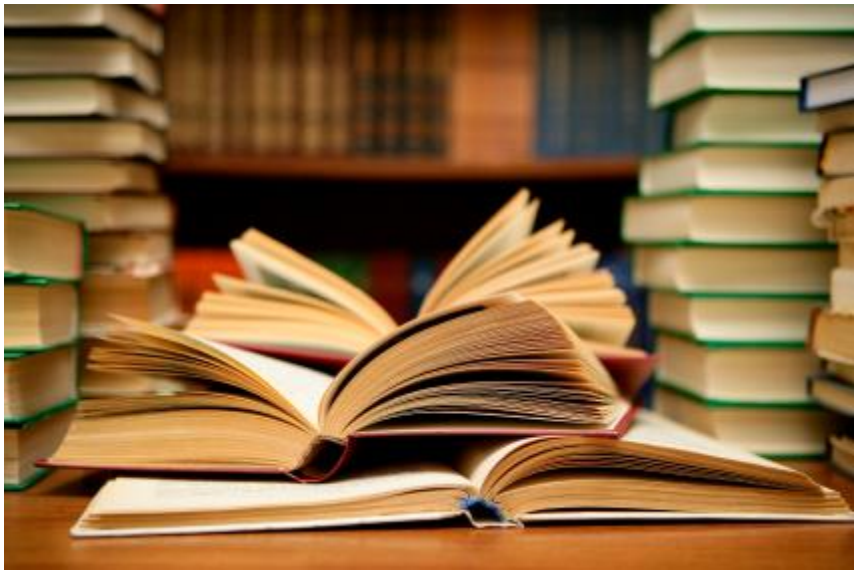


High-Level Architecture used in Watson

Watson

- Specifics
  - 16 Terabytes of RAM
  - Can process 500 gigabytes (1 million books) per second
  - Content was stored in Watson's RAM rather than memory to be more easily accessed
  - Cost about $3 Million

Watson's sources of information



- Encyclopedias
- Dictionaries
- Thesauri
- Newswire articles
- Literary works
- Databases, taxonomies, and ontologies.
- Wikipedia articles
- And more

How Watson Works

- Receives the clues (questions) as electronic texts
- It then divides these texts into different keywords and sentence fragments and searches for statistically related phrases
- Quickly executes thousands of language analysis algorithms
- The more algorithms that find the same answer increase Watson’s confidence of his answer and it calculates whether or not to make a guess

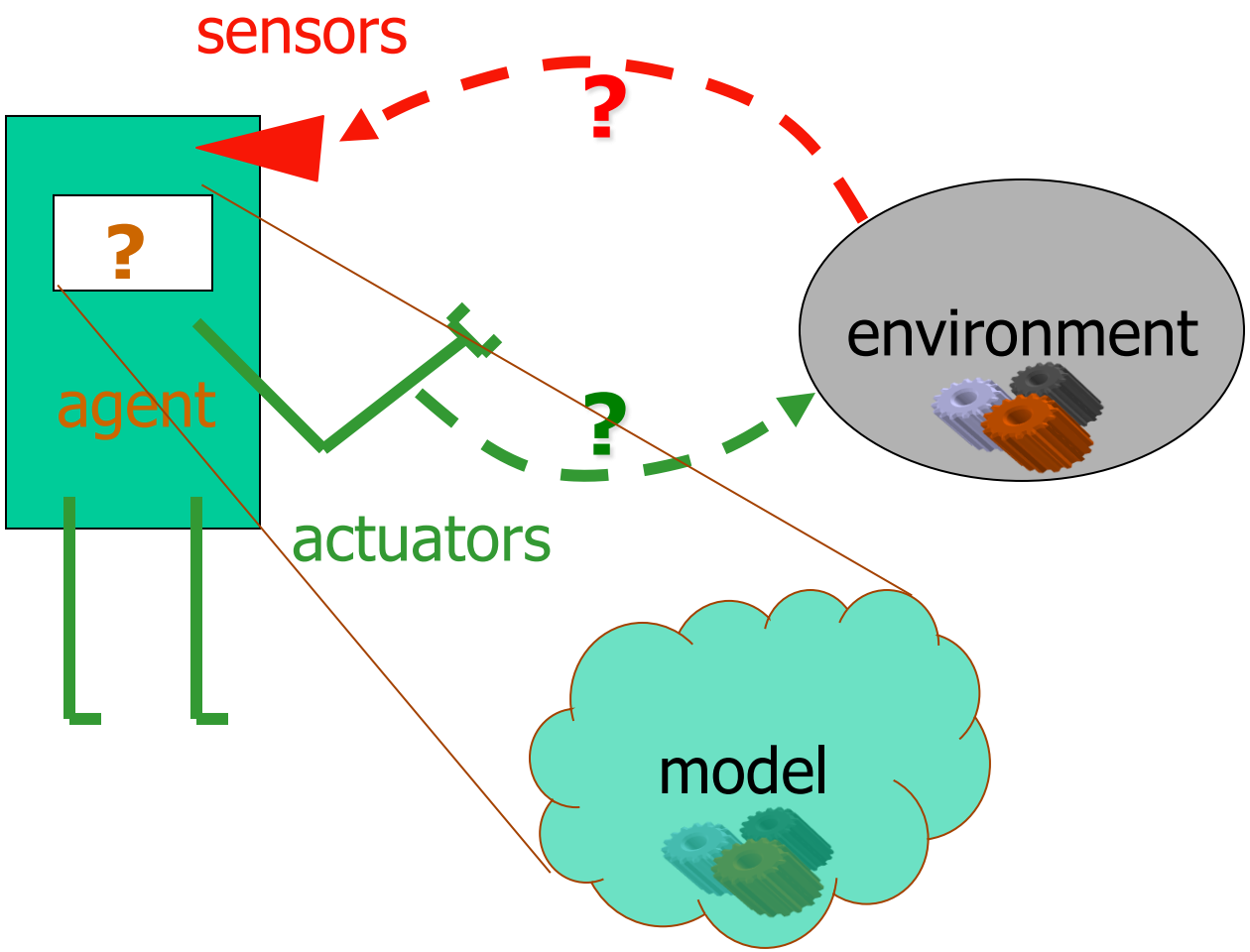
How to achieve AI?

- How is AI research and engineering done?
- AI research has both theoretical and experimental sides. The experimental side has both basic and applied aspects.
- Competitions!
- There are two main lines of research:
  - One is biological, based on the idea that since humans are intelligent, AI should study humans and imitate their psychology or physiology.
  - The other is phenomenal, based on studying and formalizing common sense facts about the world and the problems that the world presents to the achievement of goals.
- The two approaches interact to some extent, and both should eventually succeed. It is a race, but both racers seem to be walking. [**John McCarthy]**

AI competitions

- Robotics - Robocup
- Chess /other games
- Turing Test (Loebner prize)
- Theorem proving
- Planning (agent)
- Data mining
- DOD autonomous cross country driving
- Finance
- Mario AI competition
- Google AI Challenge

AI as an Agent



What is an (Intelligent) Agent?

- An over-used, over-loaded, and miss-used term.

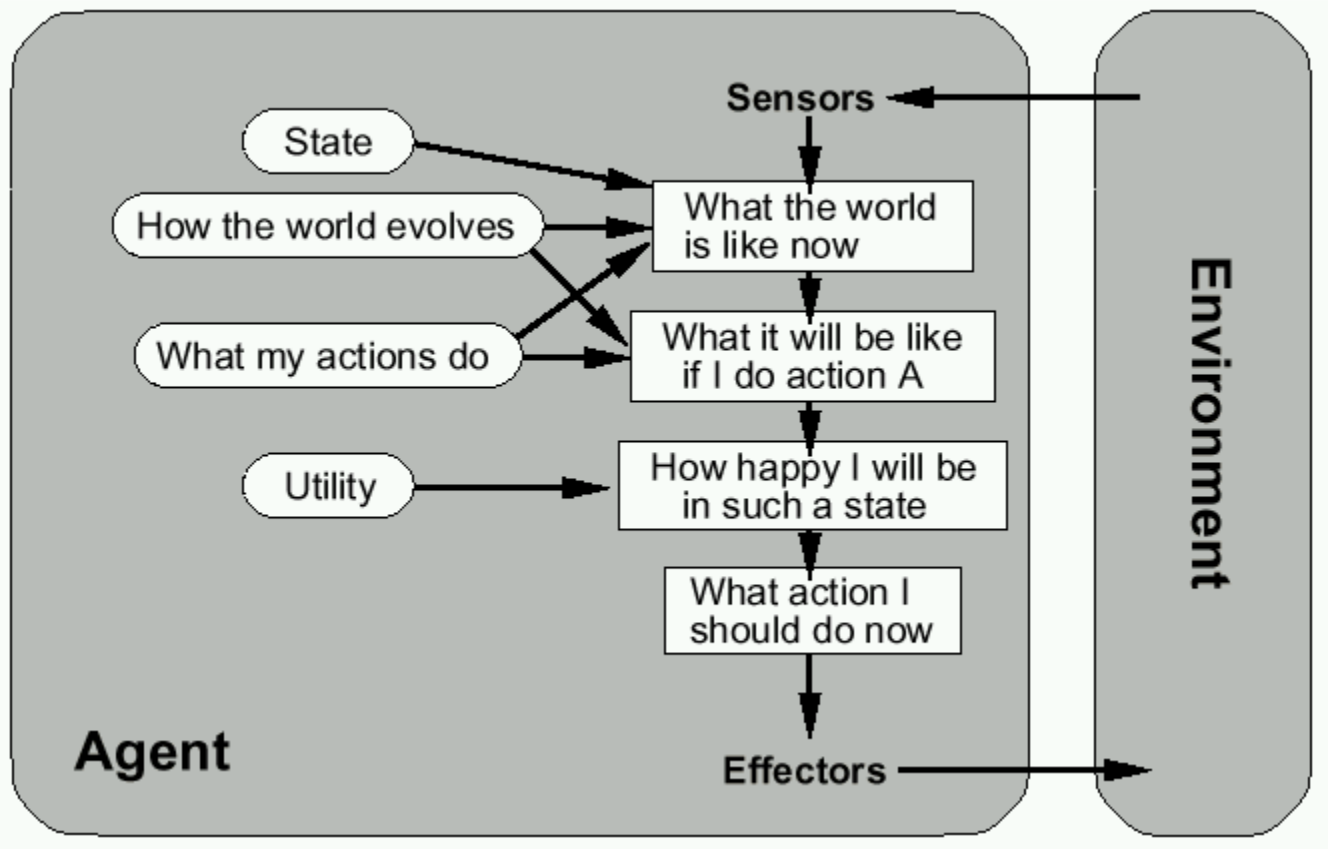
- Anything that can be *viewed as* **perceiving** its **environment** through **sensors** and **acting** upon that environment through its **effectors** to maximize progress towards its **goals**.
  - Crawlers?
  - Daemons?

- PAGE (Percepts, Actions, Goals, Environment)

- Task-specific & specialized: well-defined goals and environment

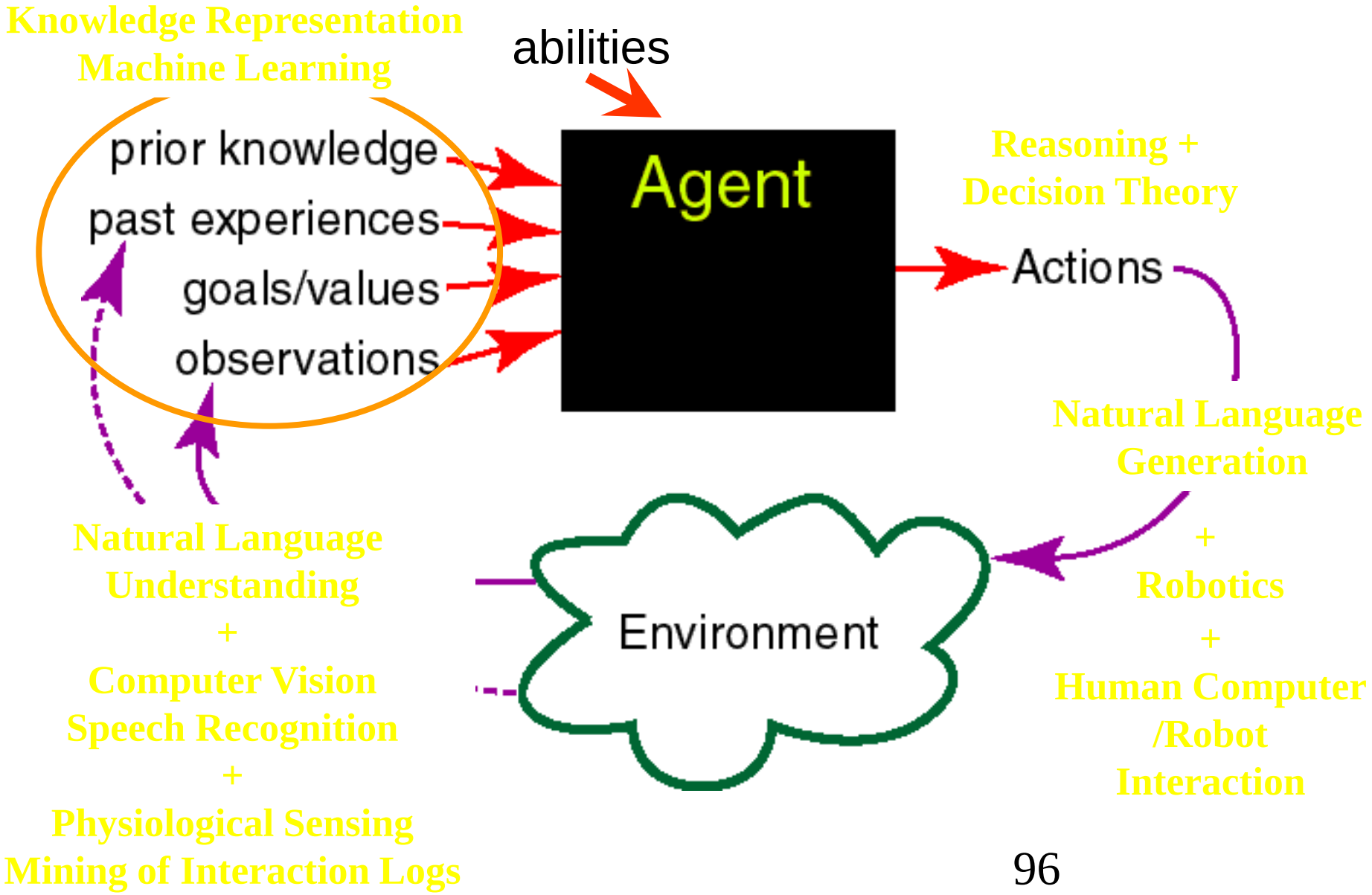
Many AI systems can be recast as Agents Systems

Agents can be quite sophisticated



Utility agent

Intelligent Agents in the World



Strong vs Weak AI

- Strong AI is artificial intelligence that matches or exceeds human intelligence — the intelligence of a machine that can successfully perform any intellectual task that a human being can.[1]
  - It is a primary goal of artificial intelligence research and an important topic for science fiction writers and futurists.
  - Strong AI is also referred to as "artificial general intelligence"[2] or as the ability to perform "general intelligent action".[3]
  - Science fiction associates strong AI with such human traits as consciousness, sentience, sapience and self-awareness.

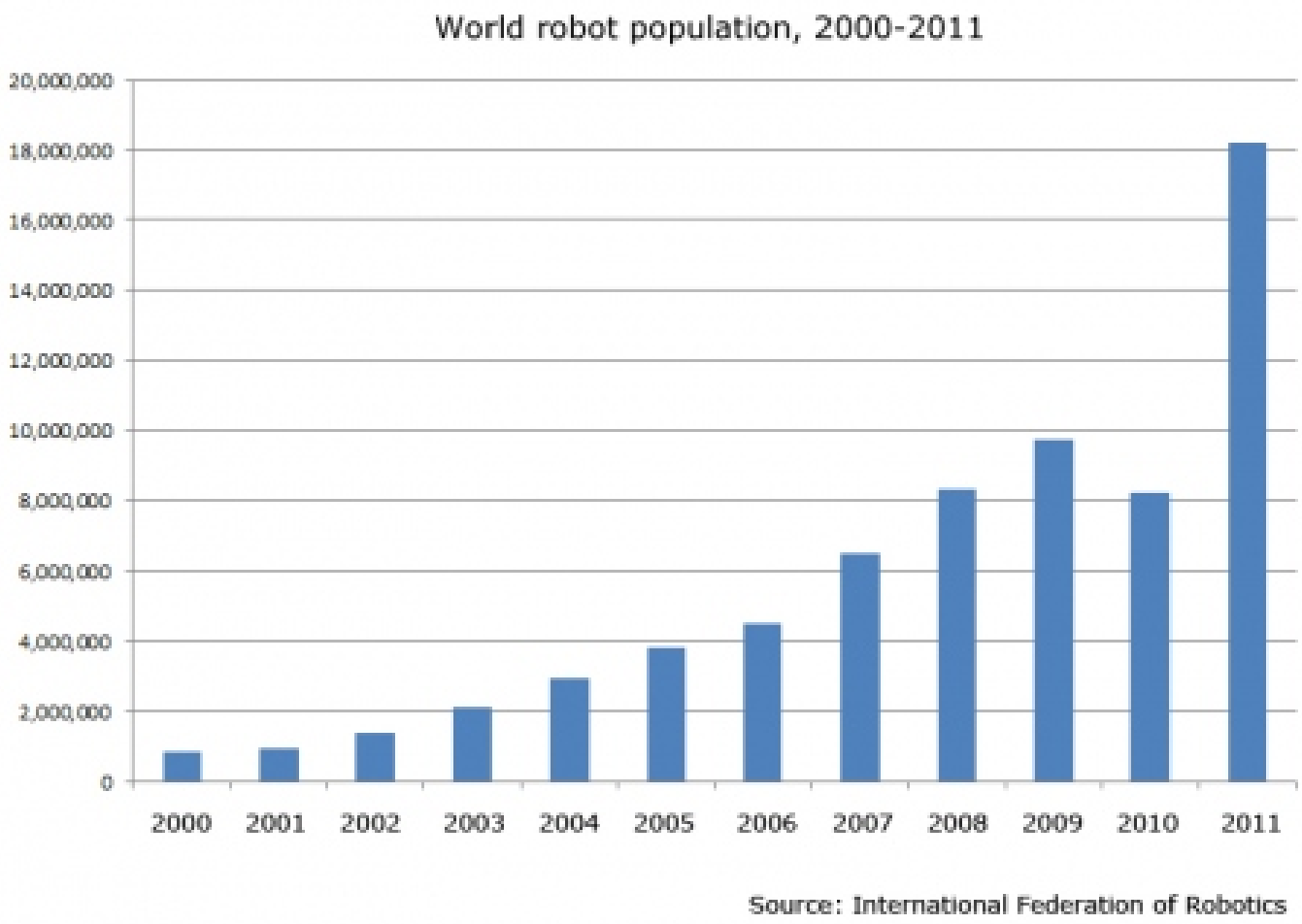
- Weak AI is an artificial intelligence system which is not intended to match or exceed the capabilities of human beings, as opposed to strong AI, which is. Also known as applied AI or narrow AI.
  - The weak AI hypothesis: the philosophical position that machines can demonstrate intelligence, but do not necessarily have a mind, mental states or consciousness. (See philosophy of artificial intelligence or John Searle's definition of Strong AI in Chinese Room)

AI State of the art - applications

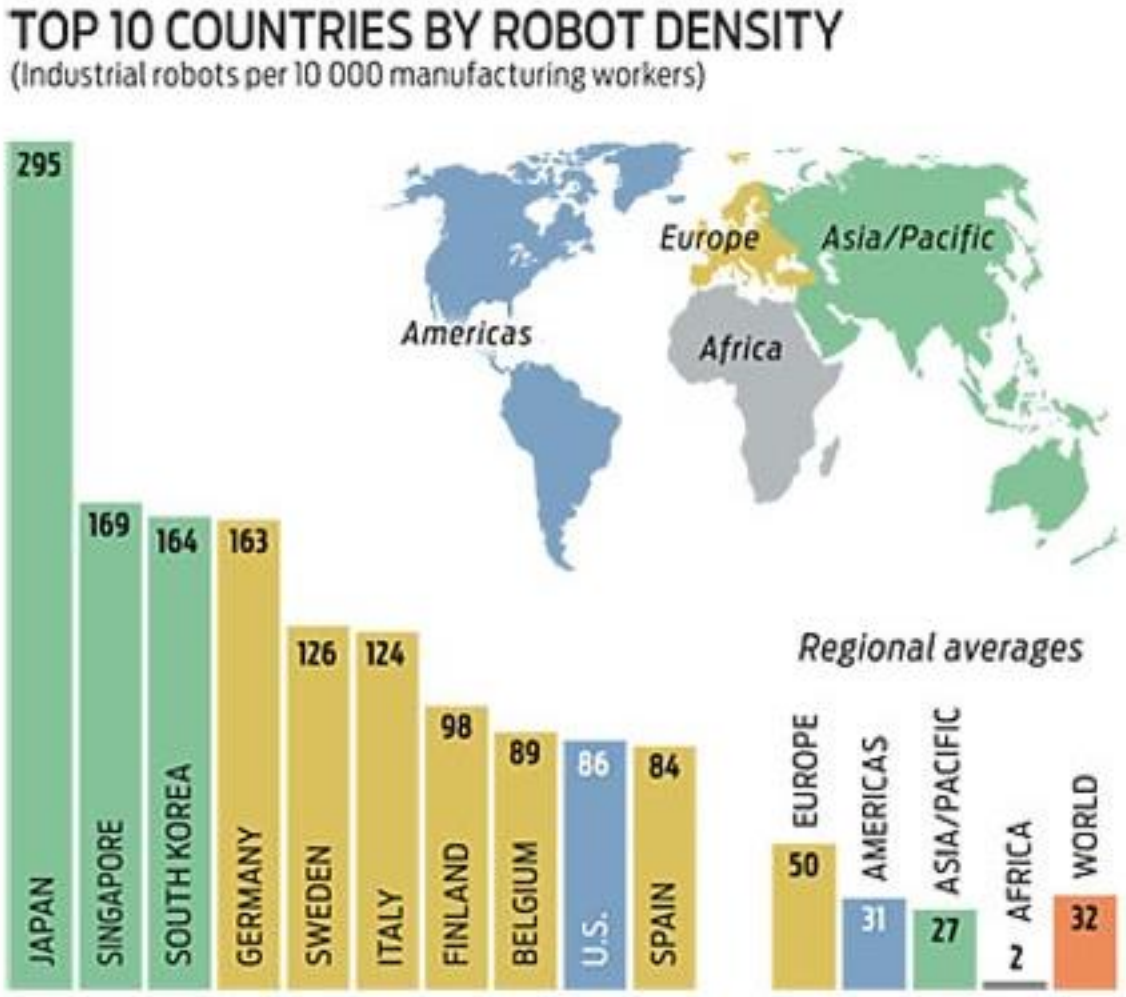
- **AI achievements:**
  - *Facilitate and replace human decision making* World-class chess and game playing
  - Robots
  - Automatic process control
  - Understand and respond to spoken language
  - Smarter search engines
  - Engage in a meaningful conversation
  - Observe and understand human emotions
  - Solving mathematical problems
  - Discover and prove mathematical theories
  - …

World robot population

The number of robots worldwide continued to grow rapidly and is now on course to reach 100 million by 2020.

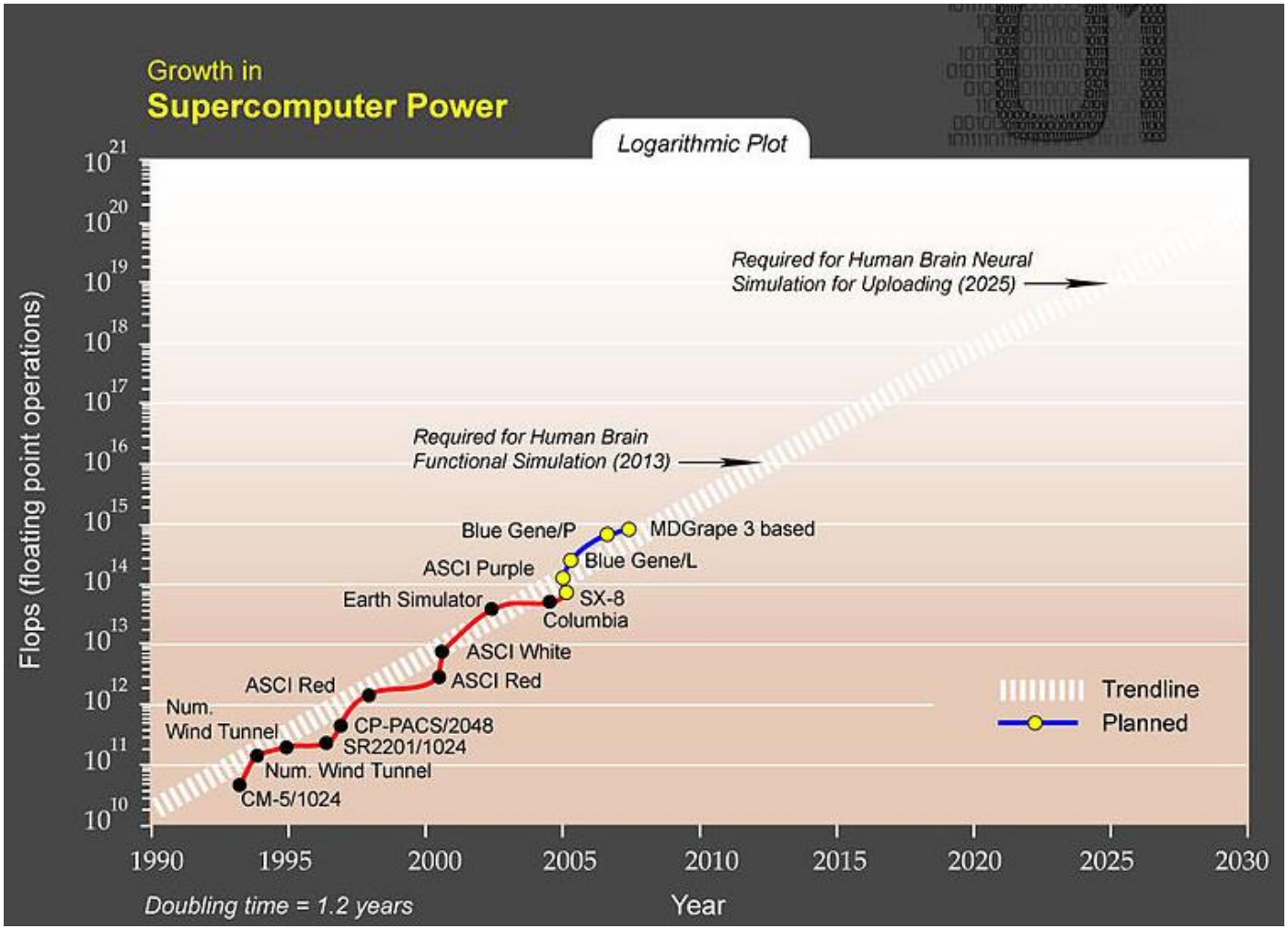


world robot population



What we know

- Applications of AI everywhere
- With Moore's law, more will appear
  - Why?



Future of AI

- Based on the continued progress of Moore's law
- Measure progress
- Brute force vs cleverness
- New apps